Spectrum similarity for Bruker NMR

This tool implements the bin-based spectrum similarity measure described in:

Lorant Bodis, Alfred Ross, Erno Pretsch, "A novel spectra similarity measure", Chemometrics and Intelligent Laboratory Systems 85 (2007) 1-8.

`spectrum_similarity.py` handles 1D spectra; `hsqc_similarity.py` extends the same method to 2D (HSQC and other processed 2D experiments). `hsqc_methods.py` adds two alternative literature methods (a Castillo-style quad-tree and a Pierens-style nearest-neighbour peak matcher) so they can be compared on the same data. `hsqc_lcc.py` adds the **Lineshape Correlation Coefficient (LCC)**, a method synthesized from all three that discriminates dense protein `1H-15N` HSQC fingerprints ~2.6× better than the previous best (see Other methods).

Method

Two spectra x and y are compared inside a common chemical-shift window. Each processed point p has a chemical shift $\delta(p)$ and an intensity $I_a(p)$ ($a \in \{x, y\}$). With the default `clip` baseline, negative intensities are zeroed first, $I_a(p) \leftarrow \max(I_a(p), 0)$.

Every point carries a weight $w(p)$ equal to its local integration element, so spectra of different digital resolution integrate consistently:

$$w_{1\mathrm{D}}(p) = |\Delta\delta(p)|, \qquad w_{2\mathrm{D}}(p) = |\Delta\delta_{F1}(p)| \cdot |\Delta\delta_{F2}(p)|.$$

Binning. At resolution n the window is split into equal bins. In 1D, a window $[\delta_{\mathrm{lo}}, \delta_{\mathrm{hi}}]$ of width $W = \delta_{\mathrm{hi}} - \delta_{\mathrm{lo}}$ gives n bins with edges $e_k = \delta_{\mathrm{lo}} + kW/n$ ($k = 0, \dots, n$). In 2D the same is done independently along $F2$ and $F1$, producing an $n \times n$ grid. The raw integral of bin k and its normalization to a target integral N_a (`--norm-x` / `--norm-y`, default 1):

$$A_k^a(n) = \sum_{p\,:\,\delta(p)\in \mathrm{bin}\ k} I_a(p)\, w(p), \qquad b_k^a(n) = N_a \frac{A_k^a(n)}{\sum_{k'} A_{k'}^a(n)}, \qquad \sum_k b_k^a(n) = N_a.$$

Similarity index. Each resolution yields a weighted Jaccard (Ruzicka) index, clamped to $[0, 1]$:

$$\mathrm{SI}_n = \frac{\sum_k \min(b_k^x, b_k^y)}{N_x + N_y - \sum_k \min(b_k^x, b_k^y)} = \frac{\sum_k \min(b_k^x, b_k^y)}{\sum_k \max(b_k^x, b_k^y)},$$

using $\sum_k \min + \sum_k \max = N_x + N_y$. In 2D the sums run over grid bins (i, j) with $b_{ij}^a(n)$ replacing $b_k^a(n)$.

The resolution runs $n = 1, \dots, N_{\mathrm{bins}}$, where the finest bins are bounded by the minimum bin width w_{min}:

$$N_{\mathrm{bins}}^{1\mathrm{D}} = \left\lfloor \frac{W}{w_{\mathrm{min}}} \right\rfloor, \qquad N_{\mathrm{bins}}^{2\mathrm{D}} = \min\left(\left\lfloor \frac{W_{F2}}{w_{\mathrm{min},F2}} \right\rfloor, \left\lfloor \frac{W_{F1}}{w_{\mathrm{min},F1}} \right\rfloor \right).$$

Upper envelope. At $n = 1$ every pair of normalized spectra collapses to a single bin, so $\mathrm{SI}_1 = 1$ is uninformative; the envelope SI_n^* smooths the descending SI_n curve. Anchors are chosen greedily: set $\mathrm{SI}_1^* = 1$; given an anchor at position m, the next anchor is the largest index $j > m$ that attains $\max_{k>m} \mathrm{SI}_k$. Between consecutive anchors SI_n^* is linear in n, and finally $\mathrm{SI}_n^* = \max(\text{interpolation}, \mathrm{SI}_n)$ so that $\mathrm{SI}_n^* \geq \mathrm{SI}_n$ everywhere.

Score. The reported similarity is the mean of the envelope over all resolutions:

$$S = \frac{1}{N_{\mathrm{bins}}} \sum_{n=1}^{N_{\mathrm{bins}}} \mathrm{SI}_n^* \ \in\ [0, 1].$$

Identical spectra give $\mathrm{SI}_n = 1$ for every n, hence $S = 1$. Because S depends on N_{bins}, only compare scores computed with the same minimum bin width.

1D

The input is a processed Bruker 1D spectrum. Pass either an experiment directory, a `pdata` directory, or a processed `1r` file. The program reads `procs` and `1r`.

Usage

```
python3 spectrum_similarity.py /path/to/exp1 /path/to/exp2
```

Useful options:

```
python3 spectrum_similarity.py exp1 exp2 --procno 1 --ppm-min 0 --ppm-max 10
python3 spectrum_similarity.py exp1 exp2 --min-bin-width 0.4 --json
python3 spectrum_similarity.py exp1 exp2 --norm-x 19 --norm-y 19
python3 spectrum_similarity.py exp1 exp2 --plot result.png
python3 spectrum_similarity.py exp1 exp2 --plot --show
```

Defaults:

- `--min-bin-width 0.4`: the value recommended in the paper for 1H NMR.
- `--baseline clip`: negative intensities are set to zero before integration.
- If no ppm range is supplied, the common ppm overlap of the two spectra is used.
- Each spectrum is normalized to total integral 1 unless `--norm-x` and `--norm-y` are supplied. For predicted or assigned proton spectra, set these to the proton counts when desired.
- `--plot` saves a graph with the two normalized spectra and the `SI_n` / `SI*_n` similarity curves. Without a path it writes `spectrum_similarity.png`.
- `--show` displays the same graph interactively when a GUI backend is available.

2D HSQC

`hsqc_similarity.py` applies the same bin-method concept to processed Bruker 2D spectra (HSQC, HMBC, COSY, …). Each spectrum is subdivided into an `n` by `n` grid of bins whose widths shrink as `n` grows, and the per-grid `SI_n` index and its `SI*_n` envelope are averaged into one score in `[0, 1]`.

```
python3 hsqc_similarity.py /path/to/exp1 /path/to/exp2
python3 hsqc_similarity.py exp1 exp2 --f2-min 0 --f2-max 10 --f1-min 0 --f1-max 160
python3 hsqc_similarity.py exp1 exp2 --min-bin-width-f2 0.1 --min-bin-width-f1 1.0
python3 hsqc_similarity.py exp1 exp2 --rotate 45                # 1H-13C HSQC
python3 hsqc_similarity.py exp1 exp2 --smooth-f2 0.05 --smooth-f1 0.5
python3 hsqc_similarity.py exp1 exp2 --plot result.png
python3 hsqc_similarity.py exp1 exp2 --json
```

The reader targets processed 2D Bruker data and reconstructs the submatrix (tile) layout:

```
experiment/
  pdata/
    1/
      2rr
      procs    # direct dimension F2 (e.g. 1H)
      proc2s   # indirect dimension F1 (e.g. 13C or 15N)
```

It reads `SI`, `OFFSET`, `SW_p`, `SF`, and `XDIM` from `procs`/`proc2s`, plus `BYTORDP`, `DTYPP`, and `NC_proc` from `procs`, to decode `2rr` and build both ppm axes.

Defaults:

- `--min-bin-width-f2 0.1`: near the 1H peak linewidth, fine enough to resolve small chemical-shift changes without dropping below the digital resolution.
- `--min-bin-width-f1 1.0`: the matching value for the 13C/15N axis, whose ppm range is roughly ten times larger. Reduce both together for more discrimination between clearly different spectra; the absolute score drops as bins shrink, so only compare scores taken at the same bin widths.
- If no ppm range is supplied for a dimension, the common overlap of the two spectra is used in that dimension.

Optional refinements (from the literature)

The core `SI_n` binning above is the exact 2D method of Bodis et al. (2009). Two paper-grounded options are available and default to **off**, because on protein `1H-15N` amide HSQC (peaks scattered, no `1H-13C` correlation) they do not help; they are intended for small-molecule `1H-13C` HSQC.

- `--rotate 45` rotates the binning grid. Bodis et al. (2009) rotate `1H-13C` HSQC spectra by 45° so the square bins cross the CHn correlation diagonal obliquely, which makes a small chemical-shift change less likely to move a peak into a neighbouring bin. For `1H-13C` HSQC this improved discrimination in the paper; on `1H-15N` data it mainly coarsens the grid.
- `--smooth-f2` / `--smooth-f1` apply a Gaussian shift tolerance (sigma in ppm) before binning, a lightweight version of the shift-insensitivity that Castillo et al. (2013) build into their tree representation. It raises the score of near-matches but also blurs genuine differences.

On the test data (base `1H-15N` HSQC vs the same-protein ligand titration and a different protein), the plain binning gave the widest separation between same-protein and different-protein spectra; rotation and smoothing narrowed it. Both options preserve a self-similarity of exactly 1.

Other methods and how they compare

`hsqc_methods.py` implements two other published HSQC similarity approaches:

```
python3 hsqc_methods.py exp1 exp2 --method quadtree   # Castillo et al. 2013
python3 hsqc_methods.py exp1 exp2 --method nn         # Pierens et al. 2012
```

- **Quad-tree (Castillo 2013)** recursively splits each spectrum at its centre of mass into a quad-tree, then compares the trees node-by-node with a similarity that blends an intensity ratio and a shift-tolerant term `alpha·min/max + (1-alpha)·exp(-gamma·d)`. The raw score is not 1 for identical spectra, so it is normalized as `s(x,y)/sqrt(s(x,x)·s(y,y))`.
- **Nearest-neighbour peaks (Pierens 2012)** picks peaks from both spectra, matches each to its nearest neighbour in the other, and maps the average peak-to-peak distance `d` to `1/(1+d)`.

`hsqc_lcc.py` implements a fourth method **synthesized from the strengths of the other three**:

```
python3 hsqc_lcc.py exp1 exp2 --f2-min 6.5 --f2-max 10 --f1-min 105 --f1-max 130
python3 hsqc_lcc.py exp1 exp2 --sigma-f2 0.03 --sigma-f1 0.30 --json
```

- **Lineshape Correlation Coefficient (LCC, this work)** renders each spectrum to one shared grid, blurs it by the physical NMR linewidth (a Gaussian per axis, `sigma = sqrt(linewidth² + expected_drift²)`), then scores the two images with the mean-centred normalized cross-correlation (Pearson / ZNCC) at **zero lag** (no shift search). The Gaussian render replaces the bin method's brittle hard edges with graded shift tolerance; mean-centring rewards *co-located* intensity and penalises intensity where the other spectrum is empty, so a differently scattered protein decorrelates instead of finding coincidental near-matches (the saturation trap of the tree and nearest-neighbour methods). Refusing to align is deliberate — alignment would let a different protein slide into registration and saturate. Self-similarity is exactly 1. Full mathematical derivation, including the graded shift-tolerance and self-similarity proofs, is in doc/LCC_method.md.

All four methods give a self-similarity of exactly 1. On the test data — a base `1H-15N` HSQC compared with the same protein plus ligand (should score high) and a *different* protein (should score low) — they separate the two cases very differently (`separation` = mean same-protein score − different-protein score; `margin` = worst same-protein score − different-protein score):

method	self	mean same	different	separation	margin
Bin (Bodis 2009)	1.00	0.79	0.49	0.29	0.24
Bin + 45° rotation	1.00	0.82	0.57	0.25	0.20
Quad-tree (Castillo 2013)	1.00	0.90	0.87	0.03	−0.01
Nearest-neighbour (Pierens 2012)	1.00	0.99	0.96	0.04	0.03
LCC (this work)	1.00	**0.94**	**0.18**	**0.75**	**0.71**

LCC separates same-protein from different-protein spectra ~2.6× better than the previous best and pushes the different protein (0.18) below *every* same-protein score. It beats the bin method across the whole physical blur range (separation 0.71–0.77 at `sigma` 0.02–0.04 / 0.20–0.40 ppm, still 0.36 even coarsened to the bin method's own resolution), so the gain is not one tuned parameter. Full numbers, robustness sweep, and a plot are in `results/`.

The quad-tree and nearest-neighbour methods were designed for **sparse small-molecule `1H-13C` HSQC** and for **shift insensitivity**: with the dense `1H-15N` amide fingerprint (150+ peaks filling one crowded region) every spectrum has a near neighbour for every peak and similar mass-centre structure, so both saturate near 1 and barely discriminate. Method choice should follow the regime: **LCC for dense protein fingerprints and titration tracking**, bins as a resolution-scanning baseline, tree/nearest-neighbour for sparse small-molecule spectra where shift tolerance matters.

Cross-regime check on `1H-13C` (sparse small molecules)

The same benchmark on **sparse small-molecule `1H-13C` HSQC** — public peak lists for six compounds each recorded twice, scoring same-compound pairs against different-compound pairs — gives the same ranking: **LCC separates best (0.81)**, bins follow (0.69), and the tree and nearest-neighbour methods saturate again (0.39 and 0.10) because over a wide window a different molecule still has a near neighbour for every peak. Their shift tolerance is real but shows up as tolerating *everything*; LCC recovers it controllably through its blur width. Run `python3.11 bench_13c.py`; details in `results/comparison_13c.md`.

References

1. L. Bodis, A. Ross, E. Pretsch, *A novel spectra similarity measure*, Chemometrics and Intelligent Laboratory Systems 85 (2007) 1-8. — the 1D bin method.
2. L. Bodis, A. Ross, J. Bodis, E. Pretsch, *Automatic compatibility tests of HSQC NMR spectra with proposed structures of chemical compounds*, Talanta 79
(2009) 1379-1386. — the 2D (`n × n` bin) extension implemented here, including the 45° rotation.
3. A.M. Castillo, L. Uribe, L. Patiny, J. Wist, *Fast and shift-insensitive similarity comparisons of NMR using a tree-representation of spectra*, Chemometrics and Intelligent Laboratory Systems 127 (2013) 1-6. — the shift-insensitive node similarity that motivates the smoothing option.
4. G.K. Pierens, S. Brossi, Z. Yang, D.C. Reutens, V. Vegh, *HSQC spectral based similarity matching of compounds using nearest neighbours and a fast discrete genetic algorithm*, Journal of Cheminformatics 4 (2012) 25. — a complementary peak-list matching approach (not bin-based).

Bruker assumptions

The reader targets processed 1D Bruker data:

```
experiment/
  pdata/
    1/
      1r
      procs
```

It uses `SI`, `OFFSET`, `SW_p`, `SF`, `BYTORDP`, `DTYPP`, and `NC_proc` from `procs` to decode the data and build the ppm axis.

Test

```
python3 -m pytest
```